

Group Theory

Binary Operation: Let G be a non-empty set. A mapping $f: G \times G \rightarrow G$ is called a binary operation on G . I.e. $\exists$ a unique $f(a,b) \in G, \forall a,b \in G$.

We denote binary operation by a symbol such as $\times, \div, +, -, \cdot, *, o$ etc.
The binary operation on G is sometimes known as binary Composition on G .

If the mapping is $* : G \times G \rightarrow G$ then we say that G is closed under operation $*$.

Ex: i) '+' is a binary operation on natural numbers N

i.e. for $1, 3 \in N \Rightarrow 1 + 3 = 4 \in N$

$\forall a, b \in N \Rightarrow a + b \in N$

(Since of the two natural numbers is always a natural number.

Ex ii) '-' is not a binary operation on $\mathbb{N}$

i.e. for $1, 3 \in \mathbb{N} \Rightarrow 1 - 3 = -2 \notin \mathbb{N}$

For $a, b \in \mathbb{N} \Rightarrow a + b \notin \mathbb{N}$

Ex. Iii) '.' is a binary operation on natural numbers $\mathbb{N}$

i.e. for $1, 3 \in \mathbb{N} \Rightarrow 1.3 = 3 \in \mathbb{N}$

Algebraic Structure: A non-empty set G equipped with one or more binary operations is called an algebraic structure. If '*' is a binary operation on G , then the algebraic structure is written as $(G, '*')$.

Ex: $(\mathbb{N}, '+')$, $(\mathbb{N}, '.')$, $(\mathbb{R}, '+')$, $(\mathbb{R}, '+, '.')$ etc

Semi- Group: An algebraic structure $(G, *)$ is said to be a semigroup if $*$ is associative in G .

(OR)

A non-empty set G , together with the binary operation $*$ is a semigroup if for $a, b, c, \in G$, $a * (b * c) = (a * b) * c$

(OR)

$(G, *)$ is said to be a semigroup if $(G, *)$ if $a * b \in G$ and $a * (b * c) = (a * b) * c$ for $a, b, c, \in G$.

Ex.1 Let N be a set of Natural No.s then $(N, +)$ is a semigroup as $a+b \in N$ for $a, b, c, \in N$

Ex.2 Let $A = \{1, 3, 5, 7, \dots\}$ be the set of odd numbers then $(A, +)$ is not a semigroup as $1+3 = 4 \notin A$.

Ex.3 Let $P(A)$ be the power set of a non-empty set A , then $(P(A), \cup)$ & $(P(A), \cap)$ are semi groups.

Monoid: A non-empty set G is said to be a Monoid with operation $*$ if it satisfies closure, associative properties and identity element exists in G .

i.e. i) $\forall a, b \in G, a * b \in G$

ii) $a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$.

iii) $\forall a \in G, \exists$ an identity 'e' $\in G \ni a * e = e * a = a$

Ex: Let Z be a set of integers then $(Z, +)$ is a monoid with 0 as the identity and $(Z, \cdot)$ is a monoid with 1 as the identity.

Ex: Let N be a set of natural nos. then $(N, +)$ is not a monoid as identity 0 is not in N

Prob: Find whether the following is a monoid or a semi group.

$S = \{a, b\}$ & $*$ $\rightarrow$ operation.

$*$	a	b
a	a	a
b	b	b

i) Closure Property: $\forall a, b \in S, a * b = a \in S$
 $\therefore S$ is closed under operation ' $*$ '.

ii) Associative Property: Let $a, b \in S$, then $a * (b * a) = a * (b) = a * b = a$,
& $(a * b) * a = (a) * a = a * = a$.
 $\therefore a * (b * a) = (a * b) * a$.
 $\therefore S$ is associative under ' $*$ '
 $\therefore (S, *)$ is a semigroup.

Let $a \in S$, then $a * b = a$
& $b * a = b$.

$\therefore a * b \neq b * a$

$\therefore$ identity element doesn't exist

$\therefore$ It is not a monoid.

$\therefore a * e = e * a = a$
then e is identity

Prob2: Show that the set $\mathbb{N}$ of natural nos is a semigroup under the operation $*$ where $x * y = \max\{x, y\}$. Is it a monoid?

Soln: Closure Property

$$\text{Let } x, y \in \mathbb{N}$$

$$x * y = \max\{x, y\}$$

$$= \text{either } x \text{ or } y \in \mathbb{N}.$$

$\therefore \mathbb{N}$ is closed under operation ' $*$ '.

Associative Property:

$$\text{Let } x, y, z \in \mathbb{N},$$

$$\begin{aligned} \text{Then } x * (y * z) &= x * \max\{y, z\} \\ &= \max\{x, \max\{y, z\}\} \\ &= \max\{x, y, z\}. \end{aligned}$$

$$\begin{aligned} \& \quad (x * y) * z &= \max\{x, y\} * z = \max\{\max\{x, y\}, z\} \\ &= \max\{x, y, z\}. \end{aligned}$$

$$\begin{aligned} \therefore x * (y * z) &= (x * y) * z \\ \Rightarrow \mathbb{N} &\text{ is associative} \end{aligned}$$

$(\mathbb{N}, *)$ is semigroup.

Identity property: Let $x \in \mathbb{N}$ then $x * 1 = \max\{x, 1\} = x$.

$$1 * x = \max\{1, x\} = x.$$

$\therefore$, 1 is the identity element

$\therefore$, $(\mathbb{N}, *)$ is a Monoid.

prob 3:- S.T. the set of all positive rational numbers forms a Monoid under the composition $\circ$ defined by $a \circ b = \frac{ab}{4}$.

Sol:- Closure property:- Let $a, b \in \mathbb{Q}^+$ (+ve rational no.s)

$$\text{Then } a \circ b = \frac{ab}{4} \in \mathbb{Q}^+$$

$$\Rightarrow a \circ b \in \mathbb{Q}^+$$

$\Rightarrow \mathbb{Q}^+$ is closed w.r.t operation ' $\circ$ '

Associative Property: For $a, b, c \in \mathbb{Q}^+$

$$(a \circ b) \circ c = \left(\frac{ab}{4} \right) \circ c = \frac{\frac{ab}{4} \cdot c}{4} = \frac{abc}{16}.$$

$$a \circ (b \circ c) = a \circ \left(\frac{bc}{4} \right) = \frac{a \cdot \frac{bc}{4}}{4} = \frac{abc}{16}.$$

$$\therefore a \circ (b \circ c) = (a \circ b) \circ c$$

$\Rightarrow \mathbb{Q}^+$ is associative under operation $\circ$.

Identity property: $\forall a \in \mathbb{Q}^+ \exists 4 \in \mathbb{Q}^+$ such that

$$a \circ 4 = \frac{a \cdot 4}{4} = a$$

$$4 \circ a = \frac{4 \cdot a}{4} = a$$

$\Rightarrow a \circ 4 = 4 \circ a = a \Rightarrow 4$ is the identity element.

$\therefore (\mathbb{Q}^+, \circ)$ is a Monoid.

Group: A non-empty set G equipped with a binary operation $*$ is said to be a group if it satisfies following properties

- 1. Closure:** $\forall a, b \in G, a * b \in G$ then closed under operation $*$.
- 2. Associative:** $\forall a, b, c \in G, a * (b * c) = (a * b) * c$, then associative w.r.t ' $*$ '
- 3. Identity:** $\forall a \in G \exists$ an identity element $e \in G \rightarrow$
 $a * e = e * a = a$. then identity property is satisfied.
- 4. Inverse:** To each $a \in G$ if $\exists$ an element $b \in G \rightarrow$
 $a * b = b * a = e$,
then ' b ' is called inverse of a & ' a ' is called inverse of b .

Abelian Group: If the group G satisfies commutative property, then it is called an abelian group.

i.e group $(G, *)$ is said to be an abelian group if $a * b = b * a$ for $a, b \in G$.

Left Cancellation Law: If a, b, c are in G then $a * b = a * c \implies b = c$

Since $a^{-1} \in G$, operating on left with a^{-1} , we get

$$a^{-1} * (a * b) = a^{-1} * (a * c).$$

$$(a^{-1} * a) * b = (a^{-1} * a) * c \quad \therefore \text{Associative property holds in } G.$$

$$\implies e * b = e * c$$

$$\implies b = c$$

Note :- For Right cancellation, operate on right with a^{-1} .

$$b * a = c * a \implies b = c.$$

Prob: Show that the set of integers w.r.t '+' forms a group. Is it abelian?

Soln: $\mathbb{Z} = \{ \dots -3, -2, -1, 0, 1, 2, 3, \dots \}$.

i) Closure property:- $\forall a, b \in \mathbb{Z}, a + b \in \mathbb{Z}$ Ex: $2 - 4 = -2 \in \mathbb{Z}$.
 $\Rightarrow \mathbb{Z}$ is closed w.r.t operation +
or $\mathbb{Z}$ satisfies closure property.

ii) Associative property:- $\forall a, b, c \in \mathbb{Z}, a + (b + c) = (a + b) + c$.
Ex: $1, -2, 3 \in \mathbb{Z}, 1 + (-2 + 3) = (1 - 2) + 3$.
 $\Rightarrow \mathbb{Z}$ satisfies Associative property.

iii) Identity property:- $\forall a \in \mathbb{Z}, \exists 0 \in \mathbb{Z} \ni a + 0 = 0 + a = a$.
 $\Rightarrow 0$ is the identity elt in $\mathbb{Z}$.
 $\Rightarrow$ identity property is satisfied.

iv) Inverse property:-
To each $a \in \mathbb{Z} \quad \exists -a \in \mathbb{Z} \quad \Rightarrow$

$$a + (-a) = (-a) + a = 0.$$

$\therefore -a$ is the inverse of a

$\Rightarrow$ Inverse property is satisfied.

$\therefore \mathbb{Z}$ is a group.

v) Commutative property :- $\forall a, b \in \mathbb{Z}, \quad a + b = b + a$
Ex: $-4, 5 \in \mathbb{Z} \quad -4 + 5 = 5 - 4$

$\Rightarrow \mathbb{Z}$ satisfies commutative property.

$\therefore \mathbb{Z}$ is an abelian group.

Prob: Prove that the set of Matrices $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, $\alpha \in \mathbb{R}$

forms a group w.r.t. Matrix multiplication if $\cos \theta = \cos \phi \Rightarrow \theta = \phi$.

Soln:- $G = \{ A_\alpha / \alpha \in \mathbb{R} \}$

Let $A_\alpha, A_\beta \in G$, where $\alpha, \beta \in \mathbb{R}$.

Closure:- $A_\alpha \cdot A_\beta = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix}$

$$= \begin{bmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -\cos \alpha \sin \beta - \sin \alpha \cos \beta \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{bmatrix}$$
$$= \begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

$\therefore \alpha, \beta \in \mathbb{R}, \Rightarrow \alpha + \beta \in \mathbb{R}$

$A_{\alpha + \beta} \in G$
 $\therefore$, Matrix mult is a binary operation.

ii) Associative: Let $A_\alpha, A_\beta, A_\gamma \in G$ where $\alpha, \beta, \gamma \in \mathbb{R}$.

Consider $A_\alpha \cdot (A_\beta \cdot A_\gamma) = A_\alpha \cdot (A_{\beta+\gamma}) = A_{\alpha+\beta+\gamma}$.

$$\& \quad (A_\alpha \cdot A_\beta) \cdot (A_\gamma) = (A_{\alpha+\beta}) \cdot A_\gamma = A_{\alpha+\beta+\gamma}.$$

$$\Rightarrow A_\alpha \cdot (A_\beta \cdot A_\gamma) = (A_\alpha \cdot A_\beta) \cdot A_\gamma$$

$\Rightarrow G$ is associative.

iii) Identity :- Let $A_\alpha \in G$ where $\alpha \in \mathbb{R}$.

$$\text{Then } \exists \underline{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = A_0 \quad \rightarrow$$

$$A_\alpha \cdot A_0 = A_0 \cdot A_\alpha = A_{\alpha+0} = A_\alpha.$$

$\Rightarrow A_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity in G .

iv) Inverse: Let $A_\alpha \in G$, then $\exists A_\beta \in G \rightarrow$

$$A_\alpha \cdot A_\beta = A_\beta \cdot A_\alpha = A_0.$$

$$A_{\alpha+\beta} = A_{\beta+\alpha} = A_0.$$

$$\Rightarrow \alpha + \beta = 0$$

$$\Rightarrow \beta = -\alpha.$$

$\therefore \alpha$ is real no., $-\alpha$ is also a real no.

$$\Rightarrow A_{-\alpha} \in G.$$

$\therefore$ Inverse property is satisfied.

$\therefore G$ is a group under Matrix multiplication.

G is not an abelian group as matrix multiplication is not commutative.

Prob3: Show that the set of all positive rational numbrs forms an abelian group under the composition defined by 'o' such that $a \circ b = \frac{ab}{3}$

Soln: Clsoure Property: $\forall a, b \in \mathbb{Q}^+$, $a \circ b \in \mathbb{Q}^+$ as $\frac{ab}{3} \in \mathbb{Q}^+$

$\therefore \mathbb{Q}^+$ satisfies closure property.

Associative property:- $\forall a, b, c \in \mathbb{Q}^+$, $a \circ (b \circ c) = a \circ \left(\frac{bc}{3}\right) = \frac{a \cdot \frac{bc}{3}}{3} = \frac{abc}{9}$.

$$\& (a \circ b) \circ c = \frac{ab}{3} \circ c = \frac{\frac{ab}{3} \cdot c}{3} = \frac{abc}{9}.$$

$$\therefore a \circ (b \circ c) = (a \circ b) \circ c$$

$\Rightarrow \mathbb{Q}^+$ satisfies Associative property.

Identity property: $\forall a \in Q^+ \overset{\text{we have to p.t.}}{\exists} e \in Q^+ \Rightarrow a \circ e = e \circ a = a.$

Consider $a \circ e = a \Rightarrow \frac{ae}{3} = a \Rightarrow e = 3.$

$\therefore \forall a \in Q^+ \exists e = 3 \in Q^+ \Rightarrow$

$$a \circ 3 = 3 \circ a = a.$$

$\therefore 3$ is the identity elt in $Q^+.$

Inverse property:- for every $a \in Q^+$ we have to p.t. $\exists b \in Q^+$

$$\Rightarrow a \circ b = b \circ a = e = 3.$$

Consider $a \circ b = 3 \Rightarrow \frac{ab}{3} = 3 \Rightarrow b = 9/a.$

$\therefore \forall a \in Q^+, \exists b = \frac{9}{a} \in Q^+ \Rightarrow a \circ \frac{9}{a} = \frac{9}{a} \circ a = 3.$

Therefore Inverse property is satisfied.

$\therefore (Q^+, 0)$ is a group.

Commutative :- $\forall a, b \in Q^+$ we have to p.T. $a \circ b = b \circ a$.

consider $a \circ b = \frac{ab}{3}$.

$$\text{Now } b \circ a = \frac{ba}{3} = \frac{ab}{3} = a \circ b.$$

$\therefore Q^+$ is commutative.

Hence $(Q^+, 0)$ is an abelian group.

Theorem: In a group, the identity element is unique.

Proof: If possible let there exists two identities e_1 and e_2 in G

Since e_1 is the identity in G

$$a.e_1=e_1.a=a, \forall a \in G \dots(1)$$

Since e_2 is the identity in G

$$a.e_2=e_2.a=a, \forall a \in G \dots(2)$$

From (1) and (2)

$$a.e_1 = a.e_2 \implies e_1 = e_2 \text{ by left cancellation law}$$

Therefore the identity element if exists, it is unique.

Theorem: In any group, the inverse of any element is unique.

Proof: If possible, let there exist two inverses b and c for the same element a where $a \in G$

Since b is the inverse of a , $ab=ba=e$ (1)

Since c is the inverse of a , $ac=ca=e$(2)

From (1) and (2)

$ab=ac \implies b = c$ (*by left cancellation law*)

Therefore the inverse if it exists it is unique.

Prob: Show that $G = \{1, -1, i, -i\}$ i.e. fourth root of unity forms a group w.r.t usual multiplication (or) S.T. the set of all roots of equation $x^4 = 1$ forms a group under multiplication.

Soln: $G = \{1, -1, i, -i\}$, operation is ' $\cdot$ '

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

$$x^4 = 1 \quad (x^2 - 1)(x^2 + 1) = 0$$

$$\Rightarrow x = \pm 1, \pm i$$

i) Closure Property: In the above table, all the elements belongs to the set G .

Therefore closure property is satisfied.

ii) From the table associative property holds good .

$$\text{i.e. } \forall a, b, c \in G, a.(b.c) = (a.b).c$$

iii) Identity Property: Since the top headed row coincides with the 1st row, headed by 1, then 1 is the identity element.

1 is the identity element under multiplication and it belongs to G.

iv) From the table 1 is the inverse of 1,

-1 is the inverse of -1

i is the inverse of -i

-i is the inverse of i

As it satisfies all the 4 properties, $(G, \cdot)$ is a group. It is also called as Klein's 4 group.

Prob: S.T. $\{1, \omega, \omega^2\}$ i.e cube root of unity forms a group w.r.t multiplication.

Note : $\omega^3=1$ and $\omega^4=\omega$

Soln: Let $G = \{1, \omega, \omega^2\}$

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

Closure property: Since all the elements in the table belongs to G , it satisfies closure property.

Associative Property: From the table , associative property holds good.

Identity property: 1 is the identity element w.r.t multiplication as top headed row coincides with the first row headed by 1.

Inverse Property: From the table 1 is the invrse of 1, ω is the inverse of ω^2 and ω^2 is the inverse of ω .

Since G satisfies all the 4 properties, $(G,.)$ is a group.

Addition Modulo 'm' ($+_m$): If a,b are any 2 integes and if m is a fixed positive integer, then a(addition modulo m)b be denoted by $a+_mb$ and is defined as the least non-negative remainder obtained when $(a+b)$ is divided by 'm'

Ex: i) $2+_35=1$ ii) $3+_5(-15)=3$ iii) $3+_52=0$

Mutiplication Modulo 'm' (X_m): If a,b are any 2 integes and if m is a fixed positive integer, then a(multiplication modulo m)b be denoted by aX_mb and is defined as the least non-negative remainder obtained when (aXb) is divided by 'm'

Prob: Prove that $G = \{0,1,2,3,4\}$ under $+_5$ forms an abelian group.

Soln: Given that $G = \{0,1,2,3,4\}$

Consider the composition table for G under $+_5$

$+_5 \backslash$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

i) closure property: Since all the elements of the table contains elements of G ,
 G is closed under $+_5$.

Associative Property: For $a, b, c \in G$.

$a +_5 (b +_5 c) \rightarrow$ the least non-negative remainder obtained when

$a + (b + c)$ is divided by 5.

$=$ least non-negative remainder obtained when $(a + b) + c$ is divided by 5.

$\therefore G$ is associative w.r.t $+_5$.

Ex: $0, 1, 2 \in G$

$$0 +_5 (1 +_5 2) = 0 +_5 (3) = 3.$$

$$\& (0 +_5 1) +_5 2 = 1 +_5 (2) = 3$$

Identity property:

Since the top headed row coincides with the first row, headed by 0, then 0 is the identity element of G under $+_5$.

Inverse Property: From the composition table it is clear that inverse of 0 is 0

Inverse of 1 is 4

Inverse of 2 is 3

Inverse of 3 is 2

Inverse of 4 is 1

Therefore inverse of every element exists in G .

Commutative property: For $a, b \in G$, we have $a +_5 b \Rightarrow$ the least non-negative remainder obtained when $a+b$ is divided by 5.
 $=$ least non-negative remainder obtained when $b+a$ is divided by 5.
 $= b +_5 a \quad \therefore$ commutative Hence $(G, +_5)$ is an abelian group.

Prob: Let $A = N \times N$ and let $*$ be a binary operation on A defined by $(a,b)*(c,d) = (ad+bc, bd)$ for all $(a,b), (c,d) \in N \times N$. Show that A has no identity element.

Soln: A is a set of all ordered pairs defined by $(a, b)*(c, d) = (ad+bc, bd)$

Clearly A is closed w.r.t operation $*$ as the result is also an ordered pair in A .

For $(a,b), (c,d), (e,f) \in A$

$$\begin{aligned} ((a,b)*(c,d))*(e,f) &= (ad+bc, bd) * (e,f) \\ &= ((ad+bc)f + (bd)e, (bd)f) \\ &= (adf + bcf + bde, bdf) \dots\dots (1) \end{aligned}$$

$$\begin{aligned} \& (a,b)*((c,d)*(e,f)) &= (a,b)*((cf+de, df)) \\ &= (a(df) + b(cf+de), b(df)) \\ &= (adf + bcf + bde, bdf) \dots\dots (2) \end{aligned}$$

From (1) & (2) clearly A is associative w.r.t $*$

Now there should exist some e such that $a * e = e * a = a$

Let $e = (0,1)$ then $((a,b) * (0,1)) = (a+0, b) = (a,b)$

& $((0,1) * (a,b)) = (0+a, b) = (a,b)$

$((a,b) * (0,1)) = ((0,1) * (a,b)) = (a,b)$

Implies $(0,1)$ is the identity element but $(0,1) \notin N$

Therefore $(0,1)$ is not the identity element in $A = N \times N$

Hence A has no identity element.

Prob: Prove that $G = \{1, 2, 3, 4\}$ under X_5 forms an abelian group.

Soln. Follow the same procedure as of addition modulo

Theorem: If every element of the group G has its own inverse, then G is abelian.

Pf: Let $a, b \in G$.

Given that $a = a^{-1}$ & $b = b^{-1}$ (1)

$\forall a, b \in G \Rightarrow a.b \in G$

$\Rightarrow a.b = (a.b)^{-1}$

$$= b^{-1}a^{-1}$$

$$= b.a$$

$\Rightarrow a.b = b.a$

Therefore G is abelian.

**Theorem: Show that if a & b are arbitrary elements of a group G then $(ab)^2 = a^2b^2$
Iff G is abelian**

Proof: Let a and b be arbitrary elements of G and suppose that $(ab)^2 = a^2b^2 \dots (1)$

$$(ab)^2 = a^2b^2 \implies (ab)(ab) = (aa)(bb)$$

$$\implies a(ba)b = a(ab)b$$

$$\implies ba = ab \text{ by cancellation laws}$$

$$\implies G \text{ satisfies commutative property}$$

Therefore G is abelian.

Now suppose that G is abelian i.e. $ab=ba \forall a, b \in G$.

$$\text{Now } (ab)^2 = (ab).(ab)$$

$$= (a(ba)b)$$

$$= (a(ab)b) \text{ Since } G \text{ is abelian}$$

$$= a^2b^2$$

Order of a group : The no. of elements in a group G is defined as the order of a group. It is denoted by $O(G)$.

Ex: $G = \{1, -1, i, -i\}$, If $(G, \cdot)$ is a group then $O(G) = 4$.

Note: A group of order ≤ 4 is an abelian group.

Subgroups: Let $(G, *)$ be a group and H be a subset of G . Then H is said to be a subgroup of G if H itself is a group w.r.t operation $*$

Note: The sets G and $\{e\}$ are called trivial or improper subgroups of G . If there exists any subgroups of G other than these 2, then they are known as proper or Non-trivial subgroups.

Theorem : The identity of a subgroup H of a group G is same as the identity of a group G .

Theorem: The inverse of any element of a subgroup H of a group G is same as the inverse of that element regarded as an element of the group G .

Theorem: A non-empty subset H of a group G is a subgroup of G iff

i) $a, b \in H \implies ab \in H$

ii) $a \in H \implies a^{-1} \in H$

Proof: Given that G is a group and $H \subseteq G$.

Suppose that H is a subgroup of a group G

It means H itself is a group.

Therefore by closure property $\forall a, b \in H \implies ab \in H$

and inverse of every element exists

So, $a \in H \implies a^{-1} \in H$.

Now Conversely suppose that

i) $a, b \in H \implies ab \in H$

ii) $a \in H \implies a^{-1} \in H$

iii) Then we have to prove that H is a subgroup of a group G .

i.e. we have to prove that H itself is a group.

i) **Closure property:** $a, b \in H \implies ab \in H$, So closure property is satisfied.

ii) **Associative Property:** As $H \subseteq G$ and G is a group, Associative law holds good.

iii) **Identity property:** Let $a \in H \implies a^{-1} \in H$ (*given*)

$$a, a^{-1} \in H \implies aa^{-1} \in H$$

$$\implies e \in H$$

Therefore identity element 'e' exists.

iv) **Inverse property:** $a \in H \implies a^{-1} \in H$

Therefore $\forall a \in H, \exists a^{-1} \in H \quad \ni aa^{-1} = a^{-1}a = e$

Therefore Inverse also exists.

Hence H is a subgroup of a group G.

Theorem: A non-empty subset H of a group G is a subgroup of a group G iff

$$ab^{-1} \in H \quad \forall a, b \in H.$$

Proof : Given that G is a group.

Suppose that H is a subgroup of a group G

Let $b \in H \Rightarrow b^{-1} \in H$ (Since H is a subgroup it will satisfy inverse property)

Now $a \in H, b^{-1} \in H \Rightarrow ab^{-1} \in H$ (by closure property)

i.e. $a, b \in H \Rightarrow ab^{-1} \in H$

Conversely suppose that $a, b \in H \Rightarrow ab^{-1} \in H$

Now we have to prove that H is a subgroup of a group G .

i) **Identity property:** Let $a \in H$, $a \in H \Rightarrow aa^{-1} \in H$
 $\Rightarrow aa^{-1} \in G$ (Since $H \subseteq G$)

But in G , aa^{-1} is denoted with e , where e is the identity element in G .

$$\Rightarrow e \in G$$

$$\Rightarrow e \in H$$

ii) **Inverse property:** $e \in H$, $a \in H \Rightarrow ea^{-1} \in H$

$$\Rightarrow a^{-1} \in H$$

$$\Rightarrow \forall a \in H, \exists a^{-1} \in H \ni aa^{-1} = a^{-1}a = e$$

iii) **Associative Property:** Since H is a subset of G , the elements of H are the elements of G

Since G satisfies Associative property, H also satisfies.

iv) **Closure Property:** $a, b \in H$

$$b \in H \Rightarrow b^{-1} \in H$$

$$a \in H, b^{-1} \in H \Rightarrow a(b^{-1})^{-1} \in H \Rightarrow ab \in H$$

Closure property is satisfied.

Hence H is a subgroup of a group G .

Theorem: The necessary and sufficient condition for a finite non-empty subset H of a group G to be a subgroup is $a, b \in H \Rightarrow ab \in H$

Proof: Suppose that H is a subgroup of G .

Then it satisfies closure property.

So $a, b \in H \Rightarrow ab \in H$

Conversely suppose that H is a finite non-empty subset and $a, b \in H \implies ab \in H$

Then we have to prove that H is a subgroup of G .

Closure property: It is already given in hypothesis.

Associative property: Since H is a finite subset of a group G , Associative property holds good.

Identity property: Given that $a, b \in H \implies ab \in H$

So $a, a \in H \implies aa \in H$

$$\implies a^2 \in H$$

$$\implies a^2 a \in H$$

$$\implies a^3 \in H$$

And $a, a^2, a^3, \dots, a^n, \dots \in H$

But given that H is finite, so some of the elements are repeated

Let $a^r = a^s$ where $r > s$

$$a^{r-s} = a^0$$

$a^{r-s} = e$ where e is the identity element

Since $r > s$ we get $r-s > 0$ or $r-s \geq 1$

$r-s$ is a positive integer and r and s are integers

$$\text{So } a^{r-s} \in H \implies e \in H$$

Therefore identity property is satisfied.

Inverse property: Since $r-s \geq 1 \implies r-s-1 \geq 0$

$$\text{Therefore } a^{r-s-1} \in H$$

Consider a . $a^{r-s-1} = a^{r-s} = e$

$$a^{r-s-1} \cdot a = a^{r-s} = e$$

a^{r-s-1} is the inverse of a . Therefore H is a subgroup of G .

Prob 1: Let $G = \{2, 4, 6, 8\}$ and the binary operation be multiplication modulo 10. Verify that $H = \{4, 6\}$ is a subgroup of group $(G, \odot_{10})$

Soln.

$\odot_{10}$	2	4	6	8
2	4	8	2	6
4	8	6	4	2
6	2	4	6	8
8	6	2	8	4

From the table clearly H is closed w.r.t $\odot_{10}$

Therefore H is a subgroup. (By the theorem, The N & S Condition for H to be a subgroup is $a, b \in H \Rightarrow ab \in H$)

Theorem: Show that the intersection of 2 subgroups of a group G is a subgroup of a group G

Proof: Let H_1 & H_2 be two subgroups of a group G

Let $a, b \in H_1 \cap H_2$

Now we have to show that $ab^{-1} \in H_1 \cap H_2$

$a, b \in H_1 \cap H_2$

$\Rightarrow a, b \in H_1$ & $a, b \in H_2$

$b \in H_1 \Rightarrow b^{-1} \in H_1$

$b \in H_2 \Rightarrow b^{-1} \in H_2$

Therefore $ab^{-1} \in H_1$ & $ab^{-1} \in H_2$

$\Rightarrow ab^{-1} \in H_1 \cap H_2$

$H_1 \cap H_2$ is also a subgroup of a group G .

Order of an element: Let $a \in G$. If n is the least positive integer such that $a^n = e$

Where e is the identity in G , then n is called the order of an element a and it is denoted by $O(a)$.

i.e. $O(a)=n$

Note. W.r.t to addition, the least positive integer n satisfying $na = e$ is called the order of an element ' a '.

If there exists no positive integer satisfying $a^n = e$ w.r.t multiplication and $na = e$ w.r.t to addition then ' a ' is said to be of infinite order or zero order.

Ex: $G = \{1, -1, i, -i\}$ and ' $\cdot$ ' is the operation.

$$a^n = e = 1$$

$$O(1) = 1 \quad (\text{since } 1^1 = 1)$$

$$O(-1) = 2 \quad (\text{since } (-1)^2 = 1)$$

$$O(i) = 4 \quad (\text{since } i^4 = 1)$$

$$O(-i) = 4 \quad (\text{since } (-i)^4 = 1)$$

Theorem: Let H be a subgroup and G be a group and $a \in G$.

Let $aHa^{-1} = \{aha^{-1} / h \in H\}$. Show that aHa^{-1} is a subgroup of G .

Proof: Consider $(ah_1a^{-1})(ah_2a^{-1})^{-1}$

$$= ah_1a^{-1}(a^{-1})^{-1}(h_2)^{-1}a^{-1}$$

$$= ah_1a^{-1}a(h_2)^{-1}a^{-1}$$

$$= ah_1e(h_2)^{-1}a^{-1}$$

$$= ah_1(h_2)^{-1}a^{-1} \in aHa^{-1} \quad \forall h_1, h_2 \in H, h_1(h_2)^{-1} \in H$$

Therefore aHa^{-1} is a subgroup of G .

Cyclic group: Let G be a group and there exist an element $a \in G$, $\exists$
 $G = \{a^n / n \in \mathbb{Z}\}$, then G is called a cyclic group .

Here 'a' is called the generator of G denoted by , $\langle a \rangle$.

Ex: $G = \{1, -1, i, -i\}$ is a cyclic group generated by integral power of i .

Theorem: Every cyclic group is Abelian.

Pf: Let G be a cyclic group. Let 'a' be the generator of G .

Let $x, y \in G$ & $x = a^n, y = a^m, m, n \in \mathbb{Z}$

$$xy = a^n a^m$$

$$= a^{n+m}$$

$$= a^{m+n}$$

$$= a^m a^n$$

$$= yx$$

$\Rightarrow G$ is abelian.

- Every subgroup of cyclic group is cyclic.
- Order of the cyclic group is same as the order of the generator.
- If 'a' is the generator of a cyclic group G, then a^{-1} is also the generator of G.

Prob: Show that the group $\langle G, * \rangle$ whose multiplication table given below is cyclic

$*$	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

From the table 'a' is the identity element of G &

$$\begin{aligned} b^2 &= b * b = c \quad ; \quad b^3 = b^2 * b = c * b = d \quad ; \quad b^4 = b^3 * b = d * b = e \quad ; \\ b^5 &= b^4 * b = e * b = f \quad , \quad b^6 = b^5 * b = f * b = a \quad . \end{aligned}$$

Thus every element of G is an integral power of 'b'.

Therefore $\langle G, * \rangle$ is a cyclic group with 'b' as the generator.

From the table, we have $f * b = b * f = a$ (identity)
 $\Rightarrow f = b^{-1}$

Hence f is also a generator of $\langle G, * \rangle$.

Prove that $\langle \mathbb{Z}_5^*, \cdot \rangle$ is a cyclic group. Find all its generators.

Soln:

$\cdot$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Now

$$2^1 = 2$$

$$2^2 = 2 \cdot 2 = 4$$

$$2^3 = 2^2 \cdot 2 = 4 \cdot 2 = 8 \equiv 3$$

$$2^4 = 2^3 \cdot 2 = 3 \cdot 2 = 6 \equiv 1$$

$$1 \cdot 1 = 1$$

$$\begin{array}{r} 5 \overline{) 10} \\ \underline{0} \\ 1 \end{array}$$

Remainder 1

$$\begin{array}{r} 5 \overline{) 61} \\ \underline{5} \\ 1 \end{array}$$

Remainder 1

It is clear that 1,2,3,4 can be expressed as an integral power of 2

Therefore 2 is the generator of this cyclic group.

Its inverse 3 is also the generator of this cyclic group.

Therefore $\langle \mathbb{Z}_5^*, \cdot \rangle$ is a cyclic group with generator s as 2 and 3

Ex: $G = \{0,1,2,3,4\}$ is a group under $+_5$

Here $na=e=0$

$O(0)=1$ (since $1 \cdot 0=0=e$)

$O(1)=5$ (since $5 \cdot 1=5 \equiv 0 = e$) Since $5 \text{div} 5$ leaving remainder 0

$O(2)=5$ (since $5 \cdot 2=10 \equiv 0 = e$) Since $5 \text{div} 10$ leaving remainder 0

$O(3)=5$ (since $5 \cdot 3=15 \equiv 0 = e$) Since $5 \text{div} 15$ leaving remainder 0

$O(4)=5$ (since $5 \cdot 4=20 \equiv 0 = e$) Since $5 \text{div} 20$ leaving remainder 0

Note: The order of every element of a finite group is finite and is less than or equal to the order of the group.

$$O(a) \leq O(G)$$

Note : In a group $O(a) = O(a^{-1})$ where a is an element in G .

Cosets: Let $(H,.)$ be a subgroup of a group $(G,.)$. Let $a \in G$, then the set $aH = \{ah/h \in H\}$ is called a left coset of H in G and $Ha = \{ha/h \in H\}$ is called a right coset of H in G generated by 'a'.

Note: w.r.t operation $+$,

$a + H = \{a + h/h \in H\}$ is called a left coset of H in G and

$H + a = \{h + a/h \in H\}$ is called a right coset of H in G generated by 'a'.

Note: If H is a subgroup of a group $(G,.)$, $h \in G$, then $h \in H$ iff $hH = H = Hh$

Note: Any two left(right) cosets of a subgroup are either disjoint or identical.

Note Let H be any subgroup of a group G , If the cosets $aH, bH, \dots$ are all disjoint then $G = H \cup aH \cup bH \cup \dots$. Where H is the coset corresponding to the identity element in G . and also $G = H \cup Ha \cup Hb \cup \dots$.

Index of H in G : If H is a subgroup of a group G , then the number of distinct left(right) cosets of H in G is called index of H in G and is denoted by $[G:H]$

Lagranges Theorem: The order of a subgroup of a finite group divides the order of the group.

Proof: Let G be a group and H be a subgroup of G .

Let $O(G) = n$ & $O(H) = m$

Let $a \in G$, then $Ha = \{ha/h \in H\}$

Any two distinct right cosets of H in G are disjoint.

Since G is a finite group, the number of distinct right cosets of H in G will be finitely say k .

Therefore $G = Ha_1 \cup Ha_2 \cup Ha_3 \cup Ha_4 \dots \cup Ha_k$

$O(G) = O(Ha_1) + O(Ha_2) + O(Ha_3) + O(Ha_4) + \dots + O(Ha_k)$

$n = m + m + m + \dots + m$ (k times)

$n = mk$

$k = n/m = O(G)/O(H)$

Therefore order of a subgroup divides order of a group.

Permutation: A one-one mapping from a set S onto itself is called a permutation.

No. of elements in set S is called degree of the permutation.

Notation: When $S = \{a_1, a_2, a_3, \dots, a_n\}$.

Let $f: S \rightarrow S$ is a permutation such that $f(a_1) = b_1, f(a_2) = b_2, f(a_3) = b_3, \dots, f(a_n) = b_n$

Where $b_1, b_2, \dots, b_n$ are nothing but the elements $a_1, a_2, a_3, \dots, a_n$ of S in some order

We write $f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix}$

Where each element in 2nd row is the f -image of the corresponding elements in the 1st row.

Identity permutation: The identity permutation on set S is defined by

$$I = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ a_1 & a_2 & a_3 & \dots & a_n \end{pmatrix} \text{ where every element is mapped onto itself.}$$

Inverse permutation: If $f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix}$ then $f^{-1} = \begin{pmatrix} b_1 & b_2 & b_3 & \dots & b_n \\ a_1 & a_2 & a_3 & \dots & a_n \end{pmatrix}$

Ex: $S = \{1, 2, 3, 4\}$

Let $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$

Then $f^{-1} = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix} \text{ after rearranging elements.}$$

Permutation Group: The set of all permutations defined over a non-empty set S forms a group, under the operation permutation multiplication. i.e. $gf(a)=g(f(a))$ and that group is called a permutation group.

- The set of all permutations defined on a set $S = \{a_1, a_2, a_3, \dots, a_n\}$ is denoted by S_n or P_n .
- We can define $n!$ bijective functions on S and all these functions are in S_n .
- Therefore order of S_n is $n!$

Composition of permutations: Let $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}$, $g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$

$$\text{Then } fog = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \quad gof = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$$

$$fog \neq gof$$

Cyclic permutation: Let $S = \{1, 2, 3, 4, 5, 6\}$

Consider the following permutation f on S

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 6 & 1 \end{pmatrix}$$

This is a cyclic permutation and can be written as $f = (1 \ 2 \ 3 \ 4 \ 5 \ 6)$

$$\text{Ex: } f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 1 & 6 \end{pmatrix}$$

$$f(1)=2, \ f(2)=3, \dots, \ f(6)=6.$$

$$= (1 \ 2 \ 3 \ 4 \ 5)(6)$$

$\therefore$ cycle of length 6.

$$\text{Ex: } f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 1 & 6 & 5 \end{pmatrix}$$

$$= (1 \ 2 \ 3 \ 4)(5, 6)$$

$\therefore$ not cyclic

Note1: A Cycle of length 1 is called identity permutation.

Note 2: Cycle of length 2 is called transposition.

Ex: I_f $S = \{1, 2, 3, 4, 5\}$ and a permutation f on S is

$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 4 & 5 \end{pmatrix}$, then $f = (2, 3)$ is a cycle of length 2 and degree 5. & $f^{-1} = (3, 2)$.

Note 3: Inverse of transposition is itself.

Note 4: Every cycle can be expressed as a product of transpositions.

Prob: write down the following products as disjoint cycles and find its inverse.

$$(1 \ 3 \ 2) (5 \ 6 \ 7) (2 \ 6 \ 1) (4 \ 5)$$

Soln: $(\overset{1}{\curvearrowright} \ 3 \ 2) (\overset{5}{\curvearrowright} \ 6 \ 7) (\overset{2}{\curvearrowright} \ 6 \ 1) (4 \ 5)$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 1 & 2 & \color{red}{4} & 6 & 7 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 6 & \color{red}{3} & 5 & 4 & 1 & \color{red}{7} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 7 & 2 & 6 & 4 & 3 & 5 \end{pmatrix}$$

$$= (1) (2 \ 7 \ 5 \ 4 \ 6 \ 3)$$

1-2-1
2-6-7
3-3-2
4-5-6
5-4-4

Since 7 is the max. no. in any cycle, we take every cycle as a permutation of degree 7 and inverse is $(1) (3 \ 6 \ 4 \ 5 \ 7 \ 2)$

Note: Every cycle can be expressed as transpositions.

Homomorphism into: Let $(G, .)$ & $(G', .)$ be two groups and $f: G \rightarrow G'$ be a function.

➤ If $\forall a, b \in G, f(a.b) = f(a).f(b)$, then f is said to be a homomorphism from G into G'

Homomorphism onto: Let $(G, .)$ & $(G', .)$ be two groups and $f: G \rightarrow G'$ be a mapping from G onto G' .

➤ If $\forall a, b \in G, f(a.b) = f(a).f(b)$, then f is said to be a homomorphism from G onto G' .

➤ It is also called a epimorphism.

Endomorphism: A homomorphism of a group G into itself is called endomorphism

Monomorphism: If the homomorphism into is one-one then it is called monomorphism.

Isomorphism: Let (G, o) & $(G', *)$ be any two groups and $f: G \rightarrow G'$ is said to be Isomorphism if

➤ $f(aob) = f(a)*f(b) \quad \forall a, b \in G$

➤ f is one-one and onto

➤ Then we say that G is isomorphic to G' and we write it as $G \equiv G'$

Ex: Let $(\mathbb{Z}, +)$ & $(G, \cdot)$ be any two groups where $\mathbb{Z}$ is the set of all integers and G is

$\{1, -1\}$. Let $f: \mathbb{Z} \rightarrow G$ defined by $f(x) = \begin{cases} 1 & \text{if } x \text{ is even} \\ -1 & \text{if } x \text{ is odd} \end{cases}$

Let us assume $3, 4 \in \mathbb{Z}$

Now $f(3+4) = f(7) = -1$

$f(3).f(4) = -1.1 = -1$

Therefore $f(3+4) = f(3).f(4)$. Therefore f is homomorphism from $\mathbb{Z} \rightarrow G$

Kernel of homomorphism: Let ϕ be a homomorphism from group G to group G'

Then kernel of ϕ is

$\ker \phi = \{x \in G / \phi(x) = e'\}$ where e' is the identity element in G'

Theorem: Let $\phi : G \rightarrow G'$ be a homomorphism of G into G' then

i) $\phi(e) = e'$ where e is the identity in G and e' is the identity in G'

ii) $\phi(a^{-1}) = (\phi(a))^{-1} \forall a \in G$

Proof: i) Consider $\phi(e) = e'$. $\phi(e)$

$$= \phi(e.e) = e'. \phi(e)$$

$$= \phi(e). \phi(e) = e'. \phi(e) \quad (\text{Since } \phi \text{ is homomorphism})$$

$$= \phi(e) = e' \quad (\text{by right cancellation law})$$

ii)

Consider $\phi(a) \cdot \phi(a^{-1})$

$$= \phi(aa^{-1})$$

$$= \phi(e)$$

$$e'$$

$$\Rightarrow \phi(a) \cdot \phi(a^{-1}) = e'$$

$$\Rightarrow \phi(a^{-1}) = (\phi(a))^{-1}$$

Theorem: The homomorphic image of a group is a group.

Proof: By def. $f(G) = \{f(a) \mid a \in G\}$

We have to prove that $f(G)$ is a group.

It is enough if we prove that $\forall a', b' \in f(G)$
 $\implies a'(b')^{-1} \in f(G)$ (By thm on subgps)

Let $a', b' \in f(G)$

$\therefore \exists a, b \in G \ni f(a) = a' \text{ \& } f(b) = b'$

Consider $a'(b')^{-1}$

$$f(a). (f(b))^{-1}$$

$$f(a).f(b^{-1})$$

$$f(ab^{-1})$$

$$\in f(G)$$

$\therefore f(G) \subseteq G' \implies f(G)$ is a subgroup of $(G', .)$

i.e. the homomorphic image of the group G is a subgroup.

i.e. the homomorphic image of a group is a group.

Theorem: Every homomorphic image of an abelian group is abelian.

Proof: Let $(G, \cdot)$ be an abelian group and $(G', \cdot)$ be a group.

$f: G \rightarrow G'$ be a homomorphism and onto

$G' = f(G)$ is the homomorphic image of G

Let $f(a), f(b) \in G'$

$$\begin{aligned} f(a) \cdot f(b) &= f(ab) = f(ba) \text{ because commutative} \\ &= f(b) \cdot f(a) \end{aligned}$$

Therefore G' is abelian.

Prob: Consider the groups $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \cdot)$. Define a function $f: \mathbb{R} \rightarrow \mathbb{R}^+$ by

$f(x) = e^x \quad \forall x \in \mathbb{R}$. Check whether these groups are isomorphic or not.

Soln: $f: \mathbb{R} \rightarrow \mathbb{R}^+$ is defined as $f(x) = e^x \quad \forall x \in \mathbb{R}$

Homomorphism: let $x, y \in \mathbb{R}$

consider $f(x+y) = e^{x+y}$

$$= e^x \cdot e^y$$

$$= f(x) \cdot f(y)$$

Therefore $f(x+y) = f(x) \cdot f(y)$

Therefore f is homomorphism.

One-one: let $x, y \in \mathbb{R}$

$$f(x) = f(y)$$

$$e^x = e^y$$

$$\Rightarrow x = y$$

Therefore f is one-one.

Onto: $\forall y \in R^+ \exists x = \log y \in R, \exists f(x) = y$

Therefore every element in R^+ has a preimage in R

Therefore $f: R \rightarrow R^+$ is isomorphism.

Normal Subgroup: Let G be a group and H be a subgroup of G .

$\forall x \in G \ \& \ h \in H$ if $xhx^{-1} \in H$ then H is said to be a normal subgroup of G denoted by $H \nabla G$.

Note: $H = \{e\}$ and $h = G$ are trivial or improper normal subgroups and if there exists any other normal subgrps, then it is called nontrivial or proper normal subgroup.

Note : A subgroup H of a group G is normal iff $xHx^{-1} = H, \forall x \in G$

Note: A subgroup H of group G is a normal subgroup of G iff each left coset of H in G is a right cose of H in G .

Theorem: Every subgroup of an abelian group is normal

Proof: Given that G is abelian group

Let $h \in H$

$$h = he$$

$$h = hxx^{-1} \text{ where } x \in G$$

$$h = (hx)x^{-1}$$

$$= (xh)x^{-1}$$

$$= xhx^{-1} \in H$$

$\Rightarrow H$ is a normal subgroup of G

Theorem: If f is a homomorphism of a group G into a group G' then the kernel of f is a normal subgroup of G .

Proof: Let $\text{Ker} f = \{x \in G \mid f(x) = e'\}$

Since $e \in \text{Ker} f$, $\text{ker} f$ is non-empty.

Let $a, b \in \text{ker} f$

Therefore $f(a)=e'$ and $f(b)=e'$

$$\begin{aligned}\text{Consider } f(ab^{-1}) &= f(a).f(b^{-1}) \\ &= f(a). (f(b))^{-1} \\ &= e'. (e')^{-1} \\ &= e'.e' \\ &= e'\end{aligned}$$

$f(ab^{-1}) = e'$
 $ab^{-1} \in \ker f$. Therefore $\ker f$ is a subgroup of G

To prove that $\ker f$ is a normal subgroup of G

let $a \in \ker f$

$$\Rightarrow f(a) = e'$$

We have to prove that $\forall x \in G, xax^{-1} \in \ker f$

$$\text{i.e. } f(xax^{-1}) = e'$$

Consider $f(xax^{-1}) = f(x).f(a).f(x^{-1})$ Since f is homomorphism

$$= f(x).f(a).f(x^{-1})$$

$$= f(x).e'.f(x^{-1})$$

$$= f(x.x^{-1})$$

$$= f(e)$$

$$f(xax^{-1}) = e'$$

$\Rightarrow \ker f$ is a normal subgroup of G

Theorem: Necessary and sufficient condition for a homomorphism f of a group G onto a group G' with kernel k to be an Isomorphism of G into G' is that $k=\{e\}$.

Proof: Necessary part: Given that f is homomorphism from G onto G'

Let f be isomorphism (one-one, onto, homomorphism)

Now we have to prove that $k=\{e\}$

Let $a \in k \implies f(a) = e'$

But $f(e) = e'$

$\implies f(a) = f(e)$

$\implies a = e$

$\implies k = \{e\}$ Since a is chosen arbitrarily

Sufficient part: Let $k=\{e\}$

We have to prove that f is one-one

i.e. we have to prove that $f(a)=f(b) \implies a=b$

Let $f(a) = f(b)$

Post multiply by $(f(b))^{-1}$

$$f(a) \cdot (f(b))^{-1} = f(b) \cdot (f(b))^{-1}$$

$$f(a \cdot b^{-1}) = f(b \cdot b^{-1})$$

$$f(a \cdot b^{-1}) = f(e)$$

$$f(a \cdot b^{-1}) = e'$$

$$\implies a \cdot b^{-1} \in k$$

But $k=\{e\}$

$$\text{Therefore } a \cdot b^{-1} = e$$

$\implies a=b$ Therefore f is one-one. f is isomorphism.